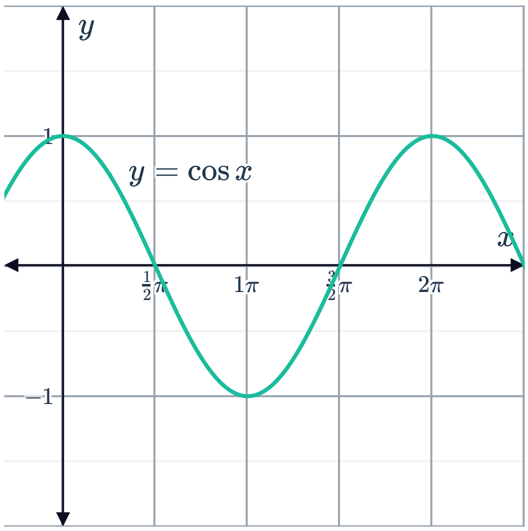
The sine and cosine functions and v[redacted]al dilations

1

a

θ	0	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	π	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	2π
$\cos\theta$	1	$\frac{1}{2}$	0	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	0	$\frac{1}{2}$	1

b



c 1

d -1

e $[-4, 4]$

2

a $(-\infty, \infty)$

b $[-1, 1]$

3

a $(-\infty, \infty)$

b $[-2, 2]$

4 8

5

a $y = 2\sin x$

b $y = -\sin x$

c $y = 3\cos x$

d $y = -1.5\cos x$

e $y = 3\sin x$

f $y = 2.5\sin x$

6 $\frac{3\pi}{2}$

7

a 8

b 0

c -8

8 5

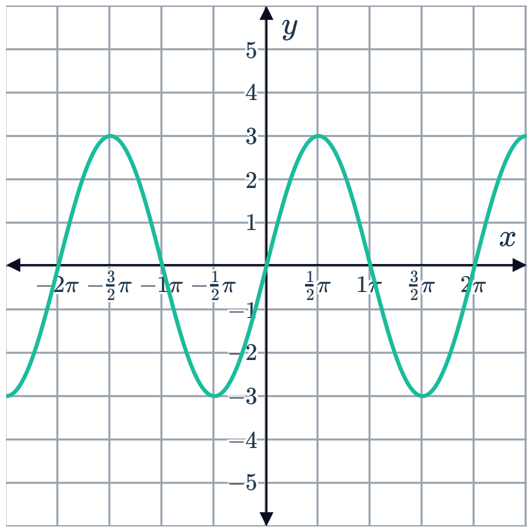
9 $-2\sin x$

10

a

i $A = 3$

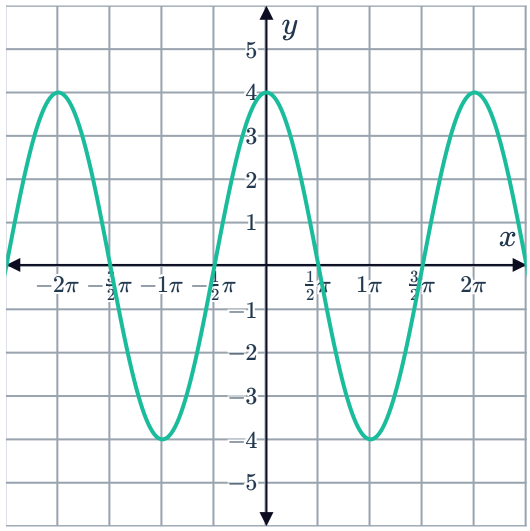
ii



b

i $A = 4$

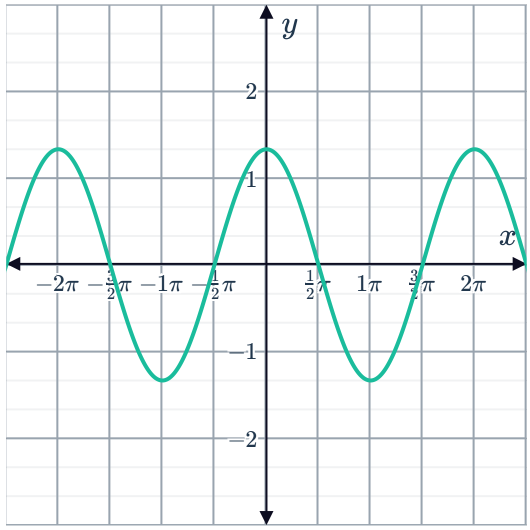
ii



c

i $A = \frac{4}{3}$

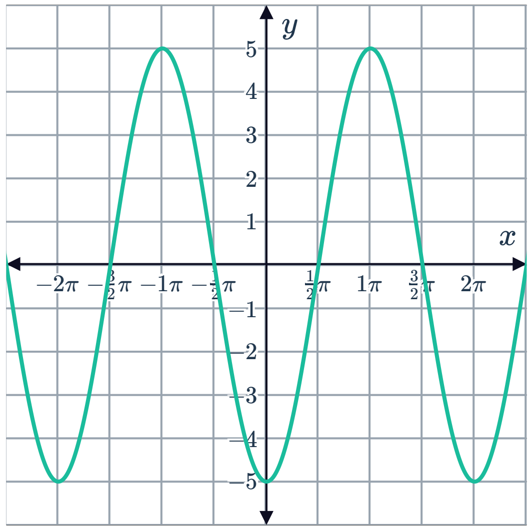
ii



d

i $A = 5$

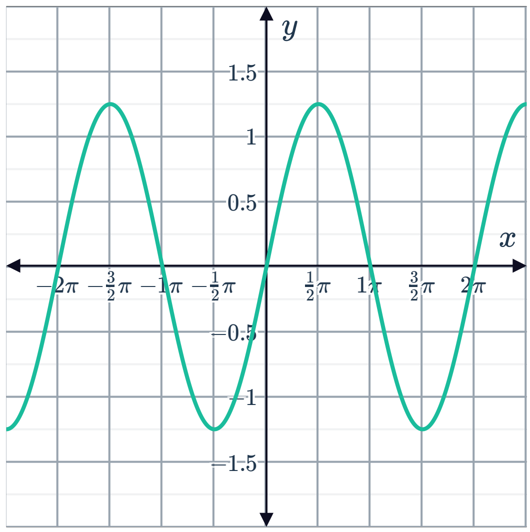
ii



e

i $\frac{5}{4}$

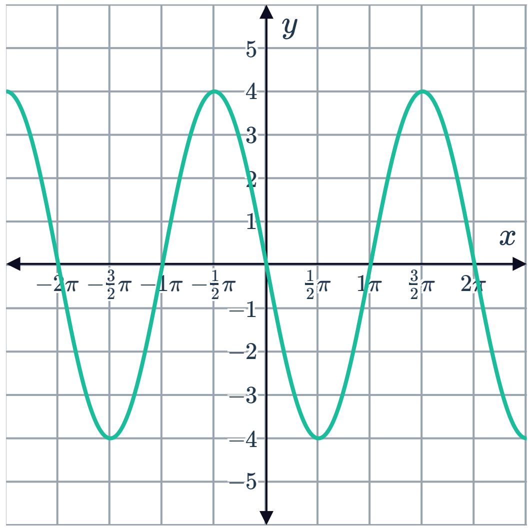
ii



f

i $A = 4$

ii



11

a 3 **b** -3 **c** 3

d A vertival dilation by a factor of 3 and reflection about the x-axis.

Vertical translations of sine and cosine functions

12 Translated 2 units downwards.

13 Translated 2 units upwards.

14 Translated 4 units upwards

15

a $y = \sin x + 4$ **b** $y = \sin x - 2$ **c** $y = \sin x - 4$ **d** $y = \sin x + 1$

16

a $y = \cos x + 3$ **b** $y = \cos x - 4$ **c** $y = \cos x - 3$ **d** $y = \cos x + 2$

17 Reflection about the x-axis, then translation 3 units upwards.

18 $y = 2$

19

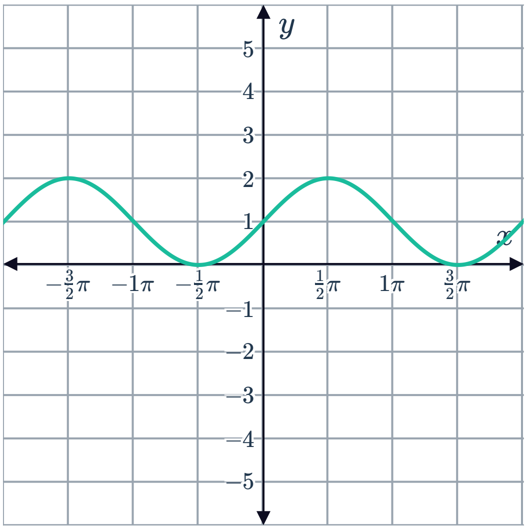
a Translated 4 units upwards. **b** 360°

20

a

i 2π **ii** 1 **iii** 2 **iv** 0

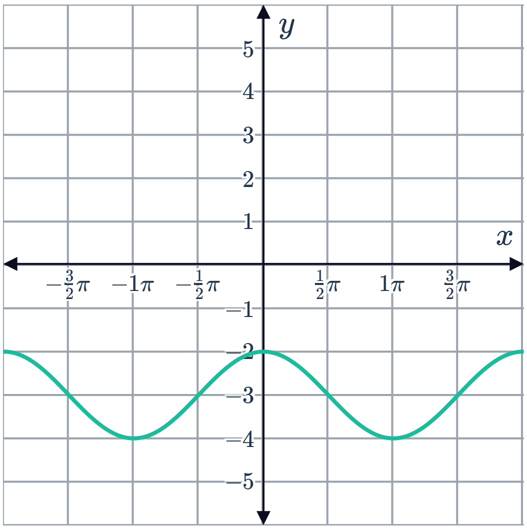
v



b

i 2π **ii** 1 **iii** -2 **iv** -4

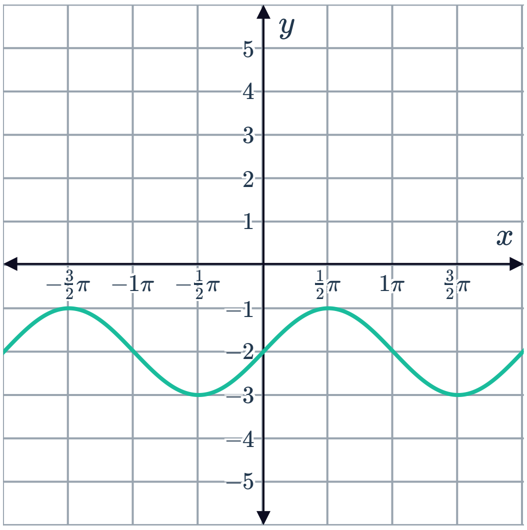
v



c

i 2π **ii** 1 **iii** -1 **iv** -3

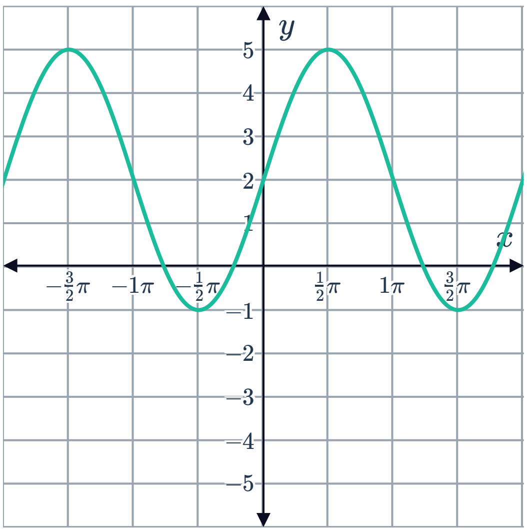
v



d

i 2π **ii** 3 **iii** 5 **iv** -1

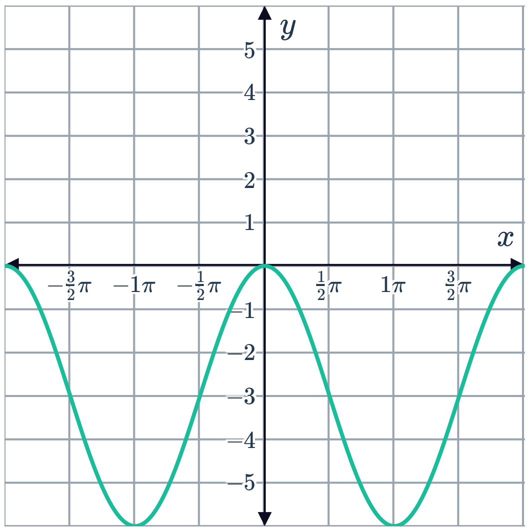
v



e

i 2π **ii** 3 **iii** 0 **iv** -6

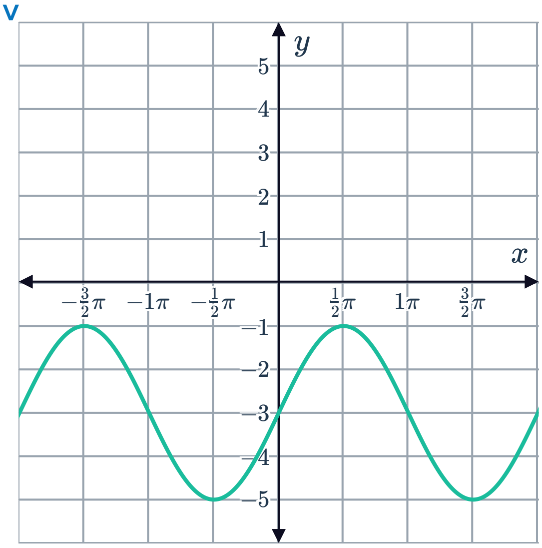
v



f

i 2π **ii** 2 **iii** -1 **iv** -5

v



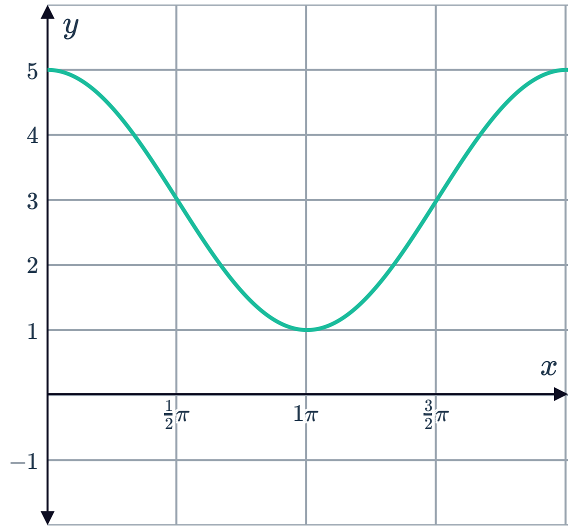
21

a

i 3 **ii** 2 **iii** 5 **iv** 1

v A vertical dilation by a factor of 2, then translated 3 units upwards.

vi

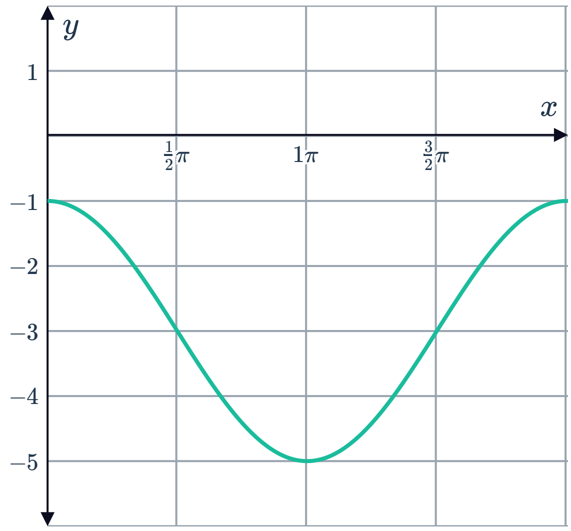


b

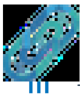
i -3 **ii** -1 **iii** -5 **iv** 2

v A vertical dilation by a factor of 2, then translated 3 units downwards.

vi

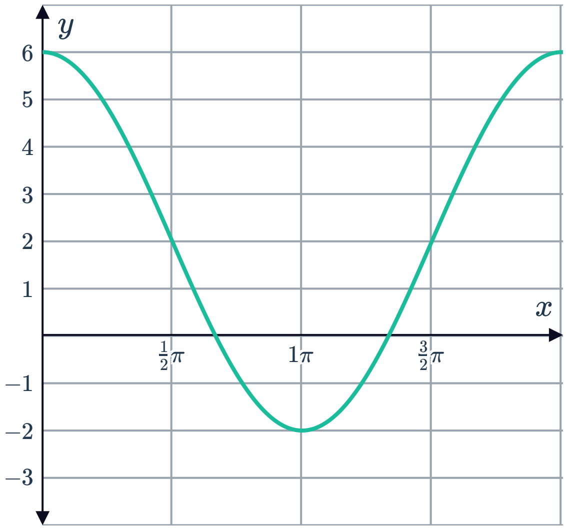


c

i 2 **ii** 6  **iii** -2 **iv** 4

v A vertical dilation by a factor of 4, a reflection about the x-axis and then translated 2 units upwards.

vi

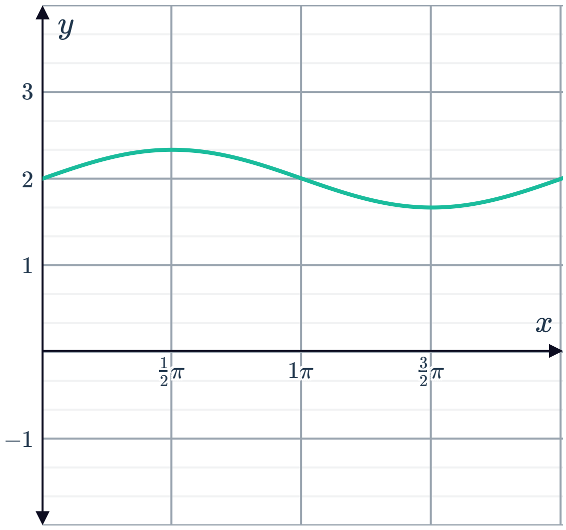


d

i 2 **ii** $\frac{7}{3}$ **iii** $\frac{5}{3}$ **iv** $\frac{1}{3}$

v A vertical dilation by a factor of $\frac{1}{3}$, then translated 2 units upwards.

vi



22

a $y = \cos x + 9$ **b** 10

23

a
- **i** 5
- **ii** -1

b
- **i** 7
- **ii** 1

c
- **i** 2
- **ii** 1

d
- **i** 2
- **ii** $\frac{1}{2}$

24 $2\sin x + 2$

25 $8\cos x - 2$

The tangent function and vertical dilations

26

a All real x except when $x = \pi k + \frac{\pi}{2}$ for all integers k.

b $(-\infty, \infty)$

c $(-\infty, \infty)$

27

a $g(x) = -3\tan x$

b $f(x) = \frac{1}{5}\tan x$

28

a
- **i** $\frac{1}{4}$
- **ii** Compressed

b
- **i** 6
- **ii** Stretched

c
- **i** $\frac{5}{3}$
- **ii** Stretched

d
- **i** 0.8
- **ii** Compressed

29

a Reflection about the x-axis or reflection about the y-axis.

b $y = -\tan x$

30

a $f(x) = 2\tan x$

b $f(x) = 0.5\tan x$

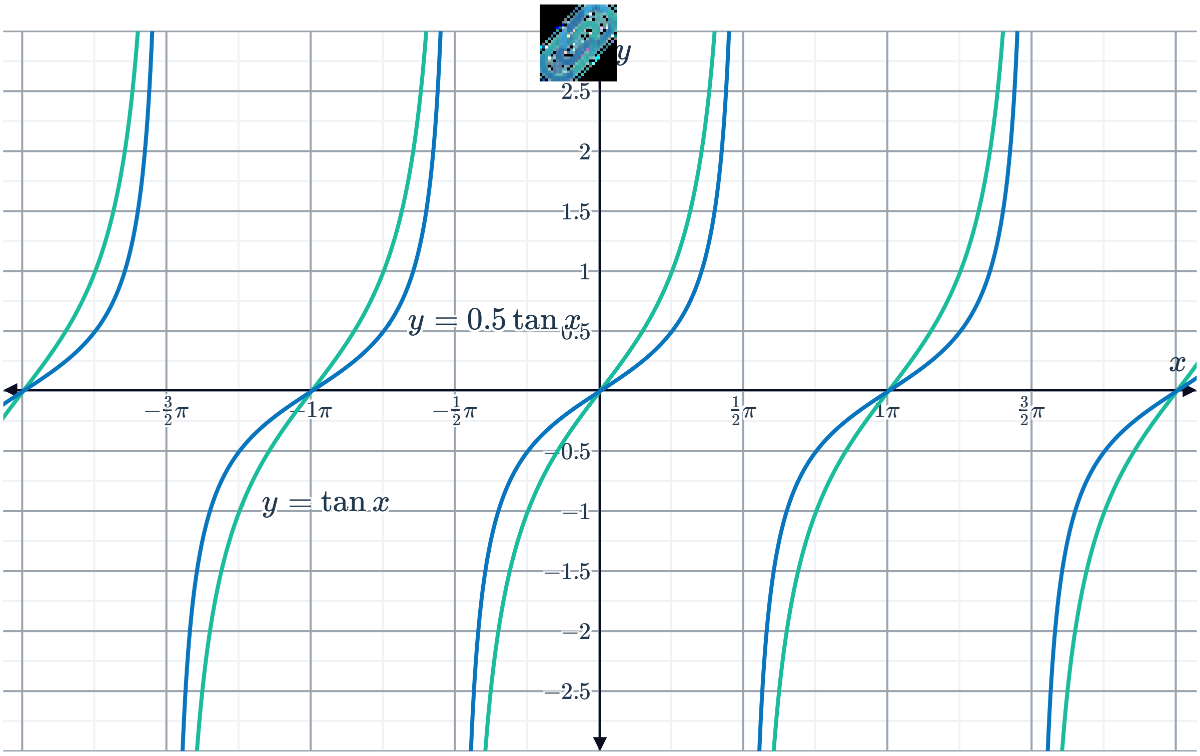
c $f(x) = 10\tan x$

d $f(x) = -2\tan x$

e $f(x) = -3\tan x$

f $f(x) = 1.5\tan x$

31



Vertical translations of the tangent function

32

a All real x except when $x = \pi k + \dfrac{\pi}{2}$ for all integers k.

b $(-\infty, \infty)$

33

a All real x except when $x = \pi k + \frac{\pi}{2}$ for all integers k.

b $(-\infty, \infty)$

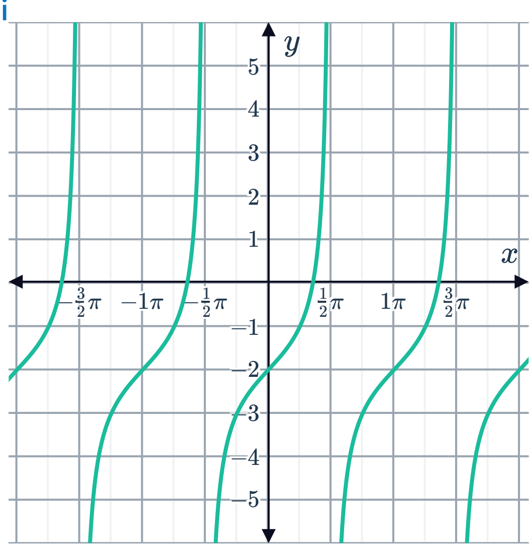
c $[3, \infty)$

34

a

i $(0,-2)$ **ii** -1 **iii** π **iv** π

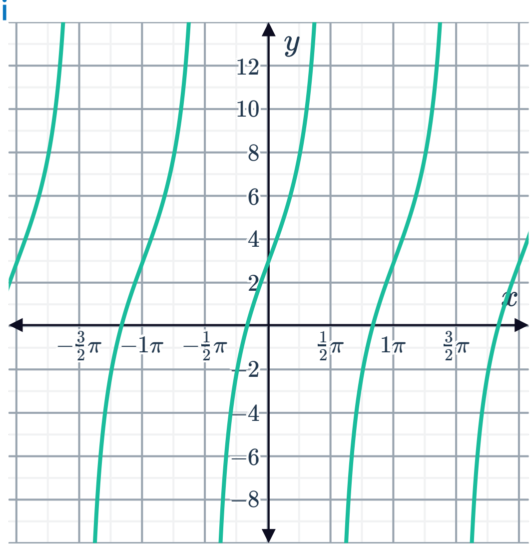
v $\dfrac{\pi}{2}$ **vi** $\dfrac{-\pi}{2}$ **vii**



b

i $(0,3)$ **ii** 8 **iii** π **iv** π

v $\dfrac{\pi}{2}$ **vi** $-\dfrac{\pi}{2}$ **vii**



35

a Translated 4 units downwards. **b** $y = \tan x - 4$